Comparative Evaluation of Two AI APIs for Multi-Object Image Recognition

Authors: Akshat Sen, Arin Lale
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Abstract

The objective of this project is to assess the performance level of both ai models, Gemini 2.0 Flash and xai/Grok 4.1 free (OpenRouter), using multi-object image recognition process. Compared to classification of an image containing a single object, the presence of multiple objects of different sizes, different colours, or objects completely or partially occluded in real-life image recognition scenarios made recognition more difficult. The chances of missed detections or hallucinated objects increased when encountering multi-object scenes rather than single-object classification. A small dataset containing both cartoon-style images and realistic photographs was collected, and for each image, the ground-truth objects were labelled. The models encountered the same prompt to both images during assessment, with the precision, recall, and F1 score being used to assess and compare the models' performance. Gemini performed best when assessing the cartoon images, while mixed results occurred when assessing the realistic images, with Grok being superior in mechanical or cluttered scenes. Both models struggled when assessing the real-world heavily cluttered images, pointing out some of the differences between each model's approach to visual complexity, as well as the current limitations of vision-enabled language models.

1. Introduction

Large multimodal models have the capacity to process data from both images and text, which opens their potential for use in tasks including automated inspection, classification systems, and understanding visual scenes. These models have demonstrated performance in processing single object tasks, however, this capability does not transfer to multiple objects tasks. Images from daily life contain a variety of items, that can differ in shape, colour, texture and light. Therefore, large multimodal models will often fail to provide accurate detections and even hallucinates additional items to be in the scene, and as a result the multi-object recognition task has the potential to evaluate the reliability of a vision model. Unlike performing a classification task, which is the identification of a single object and guessing an additional object, a multi-object recognition task requires a model to provide a complete assessment of all items that are visible. The task complexity depends on the type of image. In a cartoon style image, for example, outlines are clear, colours straightforward, but in a realistic scene there are complex textures, shadows and clutter from other objects. This is the premise of the project, comparing the performance of Gemini 2.0 Flash and Grok on multi-object recognition task using cartoon and realistic images. Each model will receive the same prompt and we will manually label the ground truth for each image_type, observing the precision, recall, F1 score of the models for patterns in performance traits and weaknesses.

2. Related Work

The study of multi-object image recognition has a significant amount of work through datasets, like MS-COCO and Open Images, which image scenes with many annotated objects. Traditional object detectors, such as YOLO and DETR, treat objects in a scene with bounding boxes and train on spatial features that allow them to output multiple object detections. In contrast, multimodal language models often rely on high-level visual embeddings and have been shown in prior work to have an unreliable approach when performing dense or cluttered scenes. Recent work has also shown that multimodal language models can be sensitive to the phrasing of the prompt and can sacrifice properly hallucinating more objects or fail to detect subtler ones. Our experiment wants to explore this direction by investigating two multimodal APIs on images that differ in visual complexity.

3. Methodology

3.1 Dataset

The dataset is composed of cartoon-like figures that display simple shapes and colours, and realistic photos of clean studio situations to cluttered ones. For each image, we manually built the ground-truth labels that include all visible objects.

3.2 Models Evaluated

1)Gemini 2.0 Flash (Google)
This multimodal approach has been developed for processing both images and text, through direct image input in combination with text prompts, and outputs object names in natural language.

2)Grok 4.1 via OpenRouter
A large language model that has been extended to include image interpretation features using the OpenRouter application programming interface (API). While it doesn't have a specific focus on object detection, it can fairly accurately list objects upon suitable prompting with an image.

3.3 Prompt Design

To Ensure fairness, both models received the same exact prompt.

"List all the objects as single words, no plural, no explanation, ignore background, separated by a comma."

This instruction encourages short and concise object names and limits descriptive sentences.

3.4 Evaluation Metrics

Because each image contains many objects simple accuracy is insufficient. We used:

- **True Positives (TP):** correctly identified objects
- **False Positives (FP):** hallucinated objects
- **False Negatives (FN):** missed objects

From these, we computed:

- **Precision = TP / (TP + FP)**
- **Recall = TP / (TP + FN)**

- **F1 Score = 2 × (Precision × Recall) / (Precision + Recall)**

These metrics reveal whether a model tends to hallucinate, miss objects, or perform consistently across both dimensions.

3.5 Implementation

The system was built on Ancient Brain using JavaScript. Users uploaded images, entered API keys, and triggered both models. Predictions were compared against ground truth, and metrics were calculated automatically.

4. Results

4.1 Summary of Performance on Cartoon and Realistic Images

Both models were tested on cartoon-style images and realistic photographs. The table below summarises the main patterns observed across precision, recall, F1 score, and error types.

Table 1 – Comparison of Gemini and Grok on Cartoon vs Realistic Images

Category	**Gemini flash2.0**	**Grok 4.1**	**Which performs better?**
Cartoon – Precision	74.71	67.06	Gemini
Cartoon – Recall	64.71	57.316	Gemini
Cartoon – F1	69.22	60.13	Gemini
Cartoon – Errors	Mostly small FN	More FN ,FP	Gemini
Realistic – Precision	63.96	67.43	Grok
Realistic – Recall	61.13	58.91	Gemini
Realistic – F1	63.39	62.08	Gemini
Realistic – Errors	More hallucinations	Fewer hallucinations	Grok
Very cluttered	Many FP + FN	Fewer FP	Grok
Clean real images	Very accurate	Good but weaker	Gemini

4.2 Cartoon Image Results

Gemini performed clearly better on cartoon style images. The simplified shapes and strong colour contrast made object boundaries easier to identify allowing Gemini to achieve higher precision and F1 scores. Grok on the other hand performed better when it came to complex cartoon samples.

4.3 Realistic Image Results

The realistic images displayed mixed performance. Gemini performed more accurately on tidally cleaned, well-lit scenes while Grok performed better on industrial or mechanical images due to fewer hallucinations.Gemini also showed more details but went out of prompt performance like signs as an object. Both models lacking heavily cluttered real photos saw large drops in accuracy with Grok being slightly more stable due to fewer false positives.

4.4 Overall Patterns

- Gemini is **consistently better on cartoon images** across all metrics.
- In Realistic images lead to **mixed results** where each model performs better under different conditions.
- Both models struggle heavily in high-clutter real environments showing the difficulty of multi object recognition.
- In high resolution examples Gemini captured more information leading to high false positive rates and not sticking to the prompt.

5. Discussion

The results suggest that visual complexity significantly impacts the performance of models. Characteristic images benefit Gemini as the simplicity of the image structure reduces ambiguity associated with visual complexity. Realistic images can include texture, lighting, and occlusion making detection more difficult; while Grok's conservative predictions enabled it to avoid hallucinating in these representations. That said, both models display obvious limitations when recognizing real-world images with varying degrees of clutter. Although the sample size of the dataset and lack of a bounding-box evaluation does not allow for thorough analysis, the results do reveal fundamentally different pattern of behavior between the two APIs.

6. Conclusion

This project involved a comparative analysis of the Gemini and Grok models during the multi-object image recognition process on both cartoon and realistic images. On cartoon images, the Gemini model consistently outperformed Grok, with results on the realistic images being mixed and contingent on the various scene types. For the cluttered real-world images, both models had a significant amount of difficulty, demonstrating that the current generation of multimodal systems still has work ahead of it in the area of exhaustive object recognition. Future work could entail a larger set of training images, the application of bounding-box metrics, and consideration of additional models.

7. References

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